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SIMULATION EXPERIMENT – 6

Design and Analysis of FIR Digital Filters Using Fourier Series and Windowing Techniques Using MATLAB

MATLAB CODE:

```
% Design and Analysis of FIR Digital Filters
% Using Fourier Series and Windowing Techniques

clc;
clear;
close all;

%% Filter Specifications
N = 30;                % Filter Order
Wn = 0.4;              % Normalized Cutoff Frequency

%% Window Functions
wr = rectwin(N+1);
wh = hamming(N+1);
whn = hanning(N+1);
wb = blackman(N+1);

%% FIR Filter Design
b_rect = fir1(N,Wn,'low',wr);
b_hamm = fir1(N,Wn,'low',wh);
b_hann = fir1(N,Wn,'low',whn);
b_black = fir1(N,Wn,'low',wb);

%% Display Filter Coefficients
```

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```
disp('Rectangular Window Coefficients');
disp(b_rect);
disp('Hamming Window Coefficients');
disp(b_hamm);
disp('Hanning Window Coefficients');
disp(b_hann);
disp('Blackman Window Coefficients');
disp(b_black);

%% Frequency Response
figure;
freqz(b_rect,1,1024);
title('Frequency Response - Rectangular Window');
figure;
freqz(b_hamm,1,1024);
title('Frequency Response - Hamming Window');
figure;
freqz(b_hann,1,1024);
title('Frequency Response - Hanning Window');
figure;
freqz(b_black,1,1024);
title('Frequency Response - Blackman Window');

%% Comparison of Magnitude Responses
[H1,w] = freqz(b_rect,1,1024);
[H2,~] = freqz(b_hamm,1,1024);
[H3,~] = freqz(b_hann,1,1024);
[H4,~] = freqz(b_black,1,1024);
```

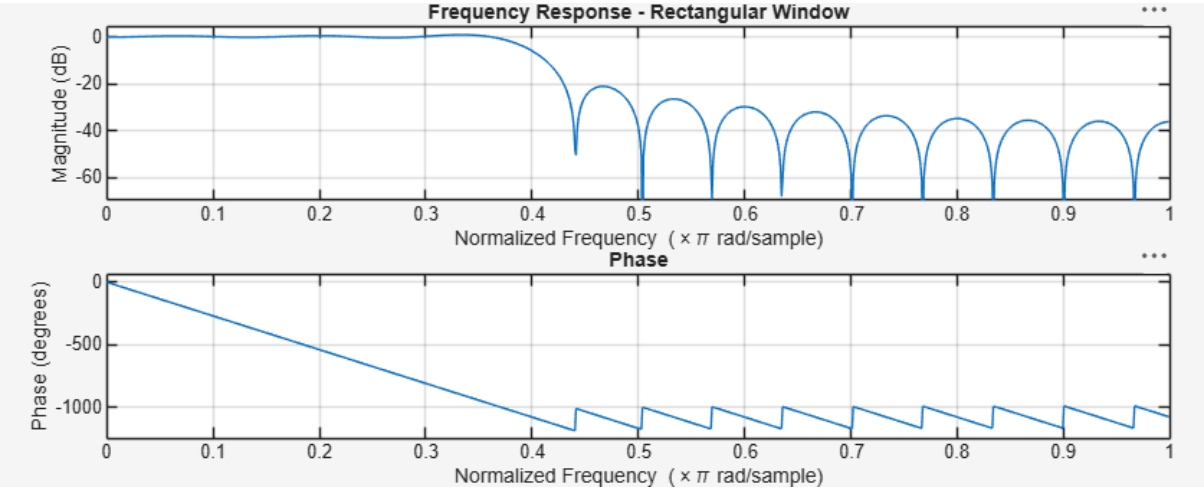
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```
figure;

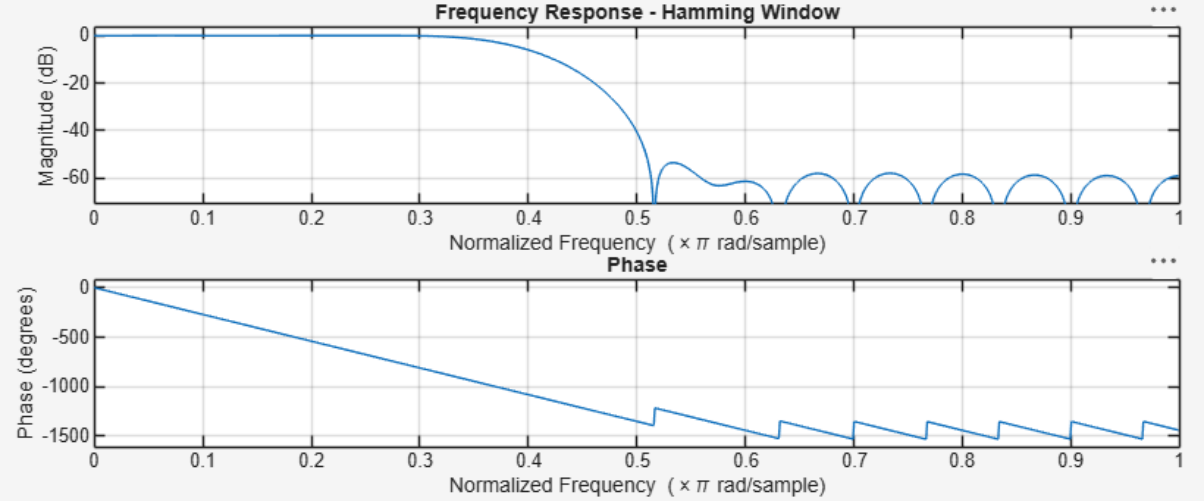

plot(w/pi,20*log10(abs(H1)),'LineWidth',1.5);
hold on;
plot(w/pi,20*log10(abs(H2)),'LineWidth',1.5);
plot(w/pi,20*log10(abs(H3)),'LineWidth',1.5);
plot(w/pi,20*log10(abs(H4)),'LineWidth',1.5);

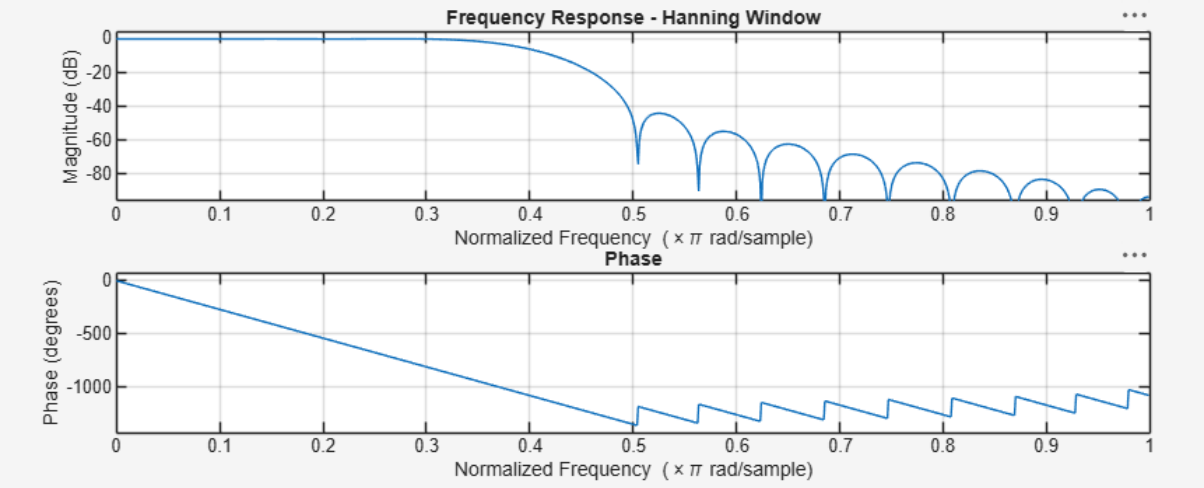

xlabel('Normalized Frequency');
ylabel('Magnitude (dB)');
title('Comparison of FIR Window Techniques');
legend('Rectangular','Hamming','Hanning','Blackman');
grid on;
```

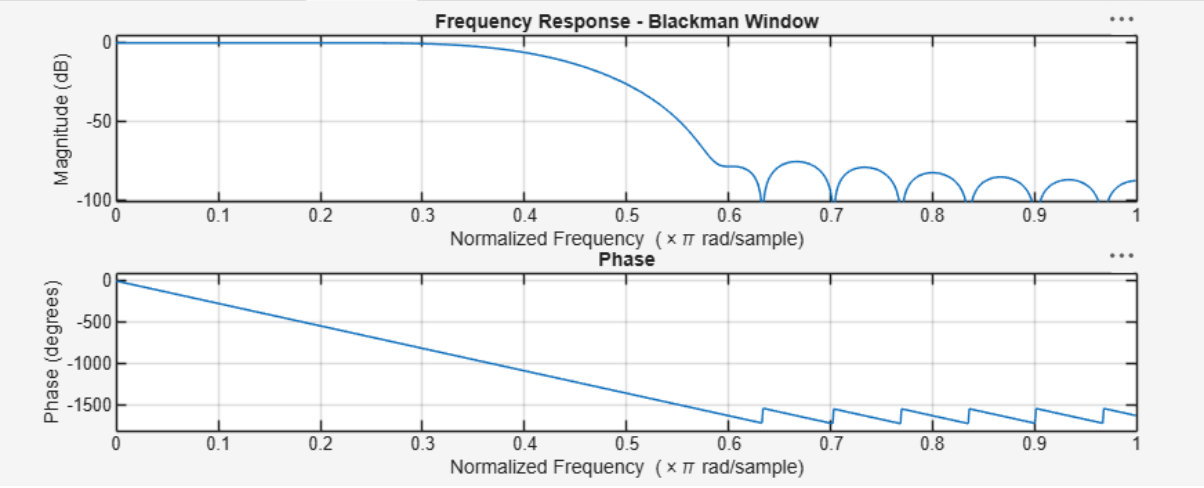
OUTPUT



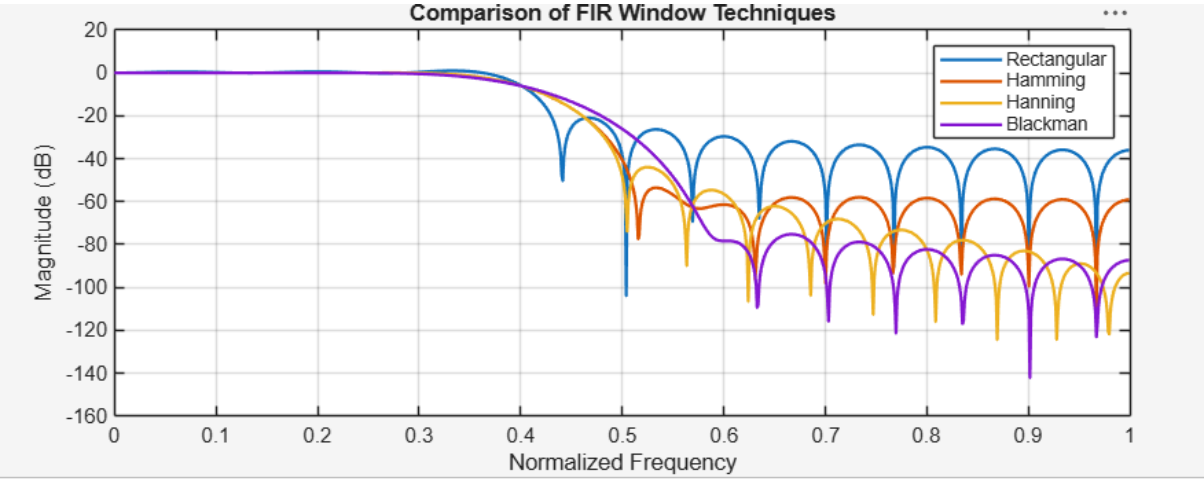
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RESULT

The FIR low-pass filter was successfully designed using **Rectangular, Hamming, Hanning, and Blackman windows** in MATLAB. The filter coefficients and frequency responses were obtained, and the results showed that different windows provide different transition widths and stopband attenuations. Thus, the performance of the various windowing techniques was successfully analyzed and compared, satisfying all the objectives and expected outcomes of the experiment.